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Class: Signals and Systems

Lab #: 05

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Lab 4: Frequency Responses

Solutions for Tasks 1–7

1. 3rd Order Elliptic Filter

```
% Given specifications
delta_p = 0.05;
delta_s = 0.05;
Wn = 0.6;

% Convert delta_p and delta_s to Rp and Rs
Rp = -20 * log10(1 - delta_p); % Passband ripple in dB
Rs = -20 * log10(delta_s);     % Stopband attenuation in dB

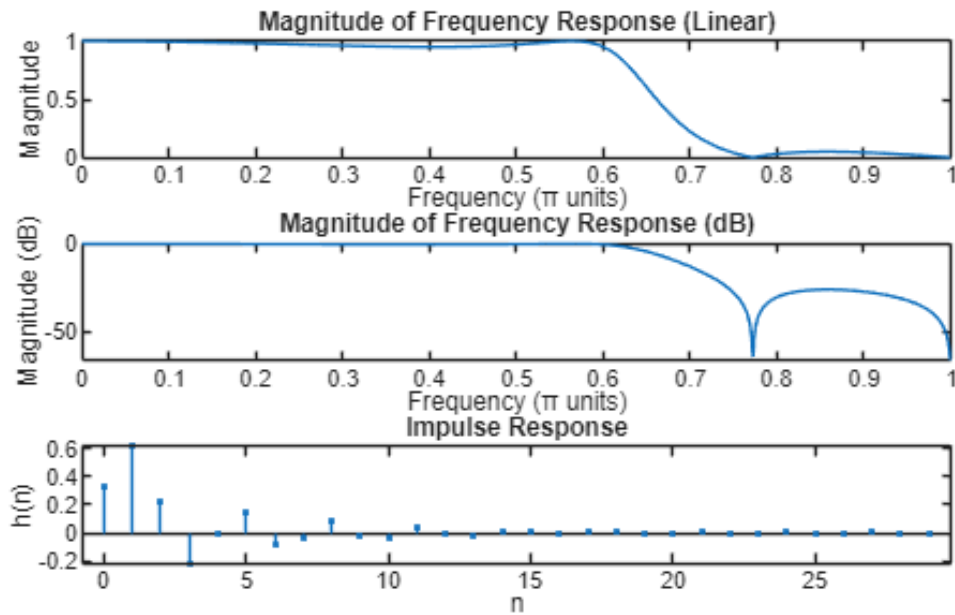
% Design a 3rd order elliptic filter
N = 3; % Filter order
[b, a] = ellip(N, Rp, Rs, Wn);

% Frequency response (magnitude)
[H, w] = freqz(b, a, 1024);
figure;
subplot(3, 1, 1);
plot(w/pi, abs(H)); % Linear scale
title('Magnitude of Frequency Response (Linear)');
xlabel('Frequency ( $\pi$  units)');
ylabel('Magnitude');

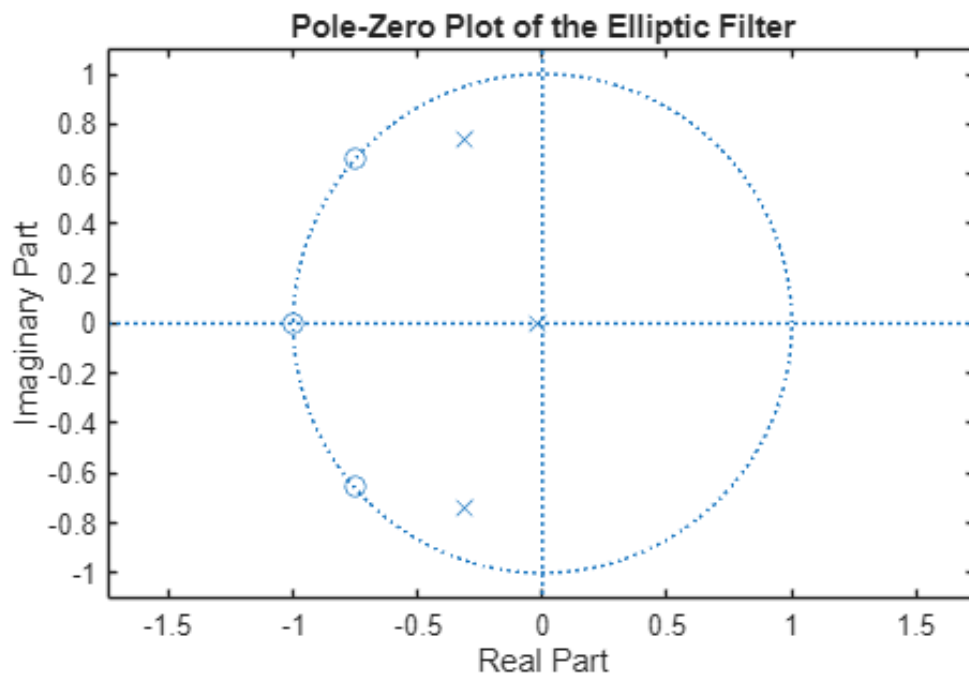
subplot(3, 1, 2);
plot(w/pi, 20*log10(abs(H))); % dB scale
title('Magnitude of Frequency Response (dB)');
xlabel('Frequency ( $\pi$  units)');
ylabel('Magnitude (dB)');

% Impulse response
h = impz(b, a, 30); % First 30 samples
subplot(3, 1, 3);
stem(0:29, h, '.');
title('Impulse Response');
```

```
xlabel('n');
ylabel('h(n)');
```



```
% Pole-Zero Plot
figure;
zplane(b, a);
title('Pole-Zero Plot of the Elliptic Filter');
```



2. 6th Order Elliptic Filter

```
% Design a 6th order elliptic filter
```

```

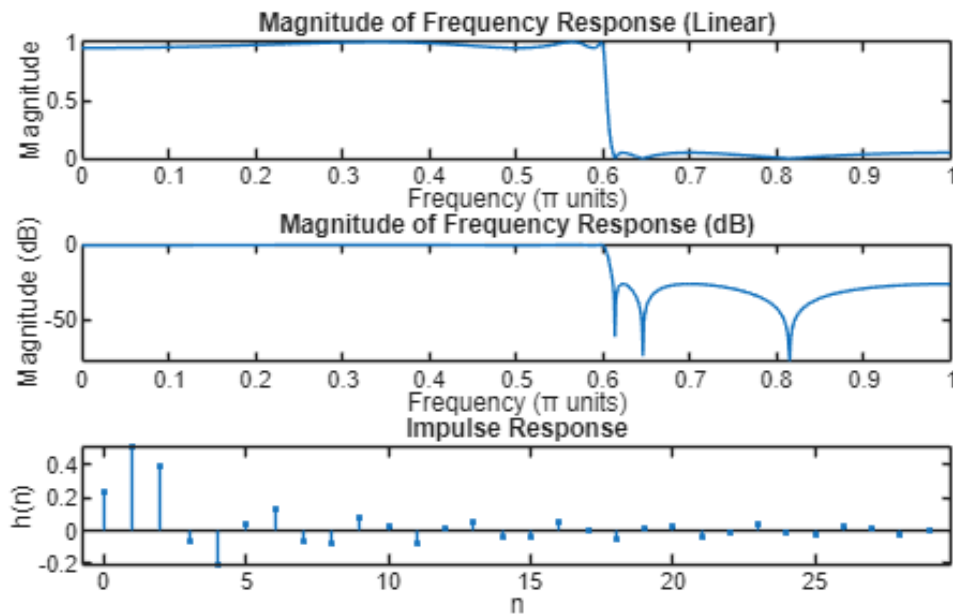
N = 6; % Filter order
[b, a] = ellip(N, Rp, Rs, Wn);

% Frequency response (magnitude)
[H, w] = freqz(b, a, 1024);
figure;
subplot(3, 1, 1);
plot(w/pi, abs(H)); % Linear scale
title('Magnitude of Frequency Response (Linear)');
xlabel('Frequency ( $\pi$  units)');
ylabel('Magnitude');

subplot(3, 1, 2);
plot(w/pi, 20*log10(abs(H))); % dB scale
title('Magnitude of Frequency Response (dB)');
xlabel('Frequency ( $\pi$  units)');
ylabel('Magnitude (dB)');

% Impulse response
h = impz(b, a, 30); % First 30 samples
subplot(3, 1, 3);
stem(0:29, h, '.');
title('Impulse Response');
xlabel('n');
ylabel('h(n)');

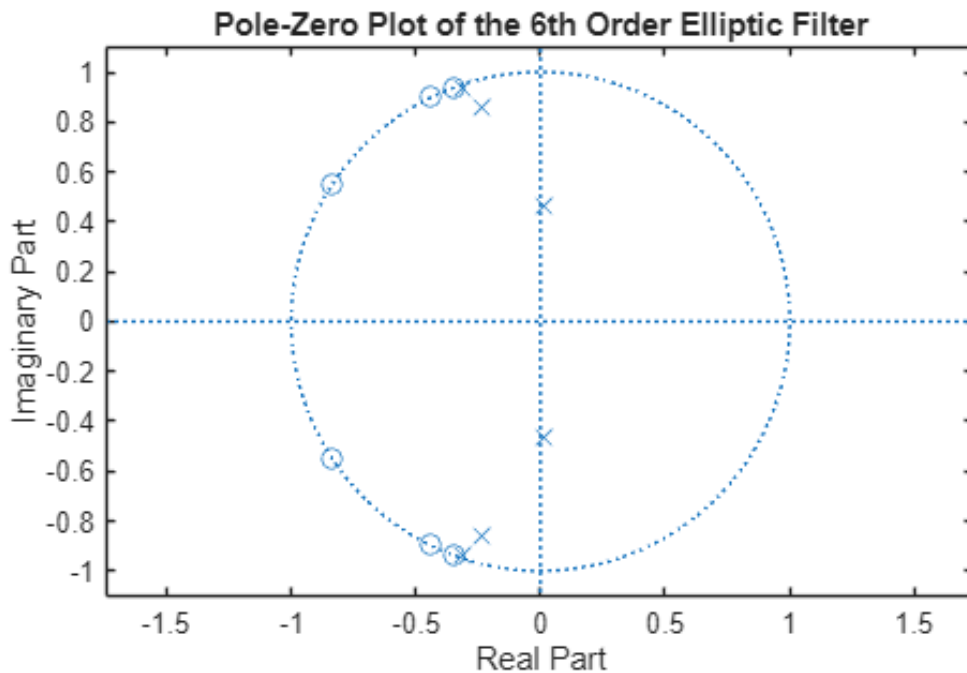
```



```

% Pole-Zero Plot
figure;
zplane(b, a);
title('Pole-Zero Plot of the 6th Order Elliptic Filter');

```



```
disp('Magnitude Frequency Response:');
```

Magnitude Frequency Response:

```
disp('3rd Order: The filter already has a sharp drop after the cut-off frequency
(at 0.6π), and you can see small ripples in the pass-band.');
```

3rd Order: The filter already has a sharp drop after the cut-off frequency (at 0.6π), and you can see small ripples

```
disp('6th Order: The drop from pass-band to stop-band is much sharper, making the
filter more selective. The transition happens more quickly.');
```

6th Order: The drop from pass-band to stop-band is much sharper, making the filter more selective. The transition happens

```
disp('Observation: Increasing the order makes the filter better at separating
frequencies, but the ripples remain .');
```

Observation: Increasing the order makes the filter better at separating frequencies, but the ripples remain .

```
disp('Impulse Response:');
```

Impulse Response:

```
disp('3rd Order: The impulse response drops off quickly and is quite clean, with
very little visible oscillations.');
```

3rd Order: The impulse response drops off quickly and is quite clean, with very little visible oscillations.

```
disp('6th Order: The impulse response still decays, but takes longer and has more
visible fluctuations or oscillations.');
```

6th Order: The impulse response still decays, but takes longer and has more visible fluctuations or oscillations.

```
disp('Observation: Higher-order filters introduce more fluctuations.');
```

Observation: Higher-order filters introduce more fluctuations.

```
disp('Pole-Zero Plot:');
```

Pole-Zero Plot:

```
disp('3rd Order: Shows 3 poles inside the unit circle and 3 zeros on the unit circle.');
```

3rd Order: Shows 3 poles inside the unit circle and 3 zeros on the unit circle.

```
disp('6th Order: Shows only 4 poles but 6 zeros, all of which lie on the unit circle.');
```

6th Order: Shows only 4 poles but 6 zeros, all of which lie on the unit circle.

```
disp('Observation: The higher-order filter doesnt just have more poles and zeros – their placement becomes more dense and the diagram looks more crowded. The extra zeros help achieve the sharper frequency cut-off.');
```

Observation: The higher-order filter doesnt just have more poles and zeros – their placement becomes more dense and

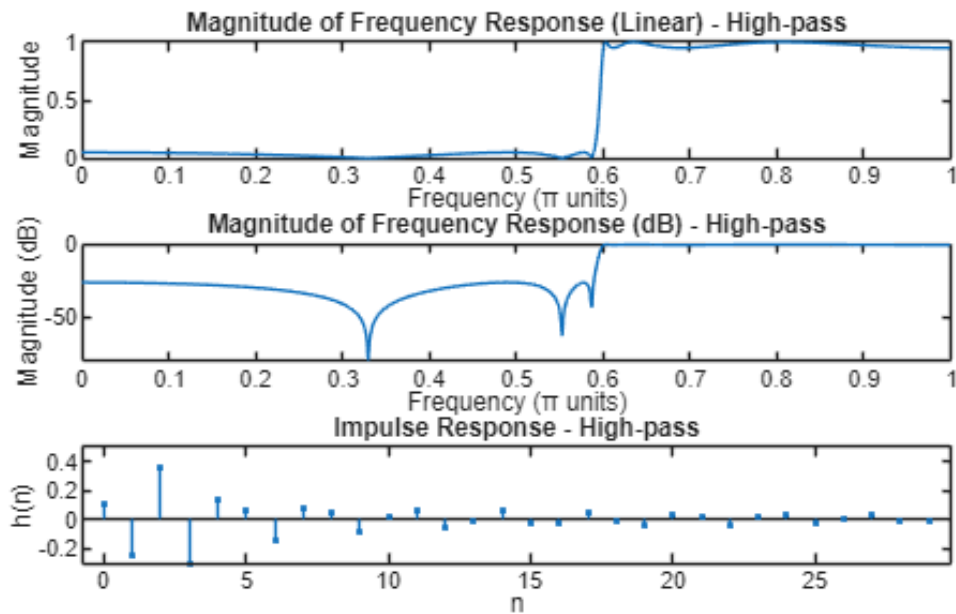
3. High-pass Elliptic Filter

```
% Design a high-pass elliptic filter
[b, a] = ellip(N, Rp, Rs, Wn, 'high');

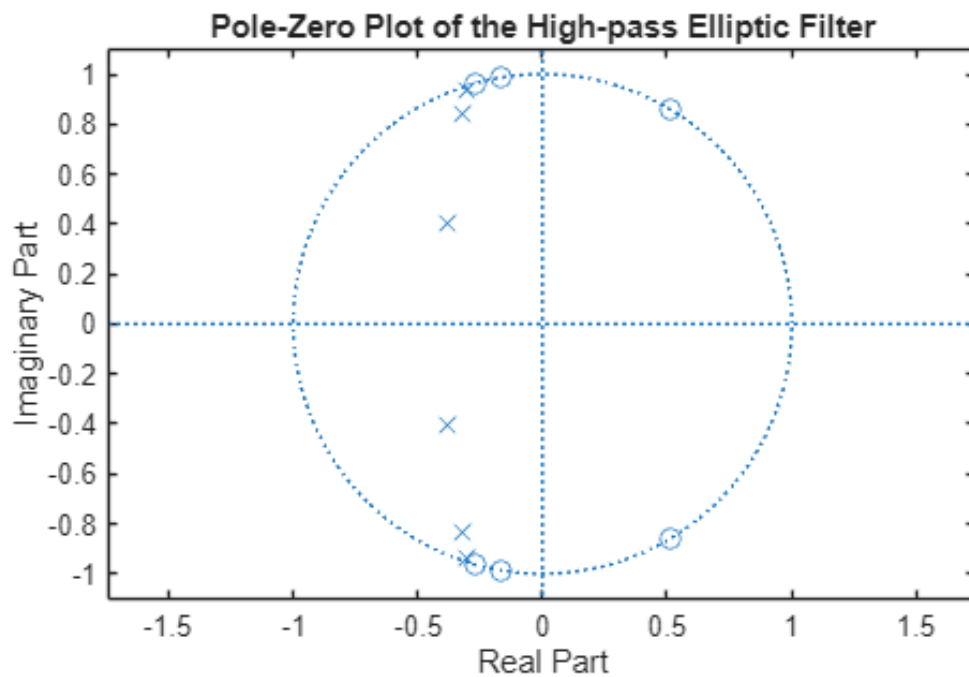
% Frequency response (magnitude)
[H, w] = freqz(b, a, 1024);
figure;
subplot(3, 1, 1);
plot(w/pi, abs(H)); % Linear scale
title('Magnitude of Frequency Response (Linear) - High-pass');
xlabel('Frequency ( $\pi$  units)');
ylabel('Magnitude');

subplot(3, 1, 2);
plot(w/pi, 20*log10(abs(H))); % dB scale
title('Magnitude of Frequency Response (dB) - High-pass');
xlabel('Frequency ( $\pi$  units)');
ylabel('Magnitude (dB)');

% Impulse response
h = impz(b, a, 30); % First 30 samples
subplot(3, 1, 3);
stem(0:29, h, '.');
title('Impulse Response - High-pass');
xlabel('n');
ylabel('h(n)');
```



```
% Pole-Zero Plot
figure;
zplane(b, a);
title('Pole-Zero Plot of the High-pass Elliptic Filter');
```



```
disp('High-Pass Elliptic Filter:');
```

High-Pass Elliptic Filter:

```
disp(' ');
```

```
disp('1. Magnitude Frequency Response:');
```

1. Magnitude Frequency Response:

```
disp('The frequency response is flipped compared to the low-pass filter.');
```

The frequency response is flipped compared to the low-pass filter.

```
disp('The pass-band starts at a higher frequency (at  $0.6\pi$ ) and stays near 1 (or 0 dB), while the stop-band is at lower frequencies.');
```

The pass-band starts at a higher frequency (at 0.6π) and stays near 1 (or 0 dB), while the stop-band is at lower frequencies.

```
disp(' ');
```

```
disp('2. Impulse Response:');
```

2. Impulse Response:

```
disp('The impulse response decays quickly, similar to the low-pass version, but with a different pattern because it's a high-pass filter.');
```

The impulse response decays quickly, similar to the low-pass version, but with a different pattern because it's a high-pass filter.

```
disp(' ');
```

```
disp('3. Pole-Zero Plot:');
```

3. Pole-Zero Plot:

```
disp('Similar to the low-pass version, but the zeros and poles are placed differently to reflect the high-pass nature.');
```

Similar to the low-pass version, but the zeros and poles are placed differently to reflect the high-pass nature.

```
disp('In the high-pass case, the poles are positioned near the unit circle in the left half of the z-plane, while the zeros are placed on the unit circle.');
```

In the high-pass case, the poles are positioned near the unit circle in the left half of the z-plane, while the zeros are placed on the unit circle.

```
disp('This configuration ensures that lower frequencies are attenuated, while higher frequencies are passed through.');
```

This configuration ensures that lower frequencies are attenuated, while higher frequencies are passed through.

```
disp(' ');
```

4. Band-pass Elliptic Filter

```
% Design a band-pass elliptic filter
Wn = [0.2, 0.6]; % Passband frequencies
[b, a] = ellip(N, Rp, Rs, Wn);
```

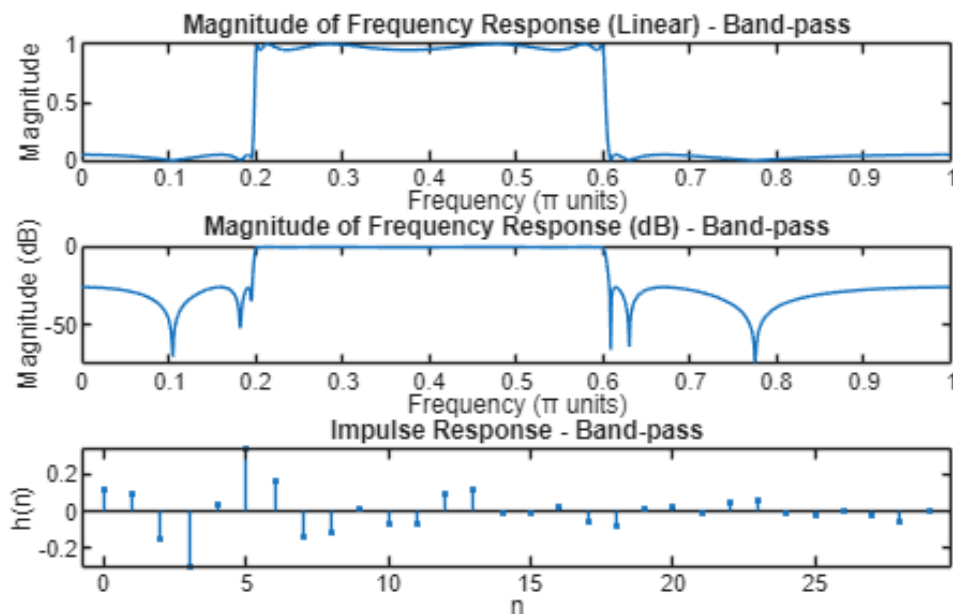
```

% Frequency response (magnitude)
[H, w] = freqz(b, a, 1024);
figure;
subplot(3, 1, 1);
plot(w/pi, abs(H)); % Linear scale
title('Magnitude of Frequency Response (Linear) - Band-pass');
xlabel('Frequency ( $\pi$  units)');
ylabel('Magnitude');

subplot(3, 1, 2);
plot(w/pi, 20*log10(abs(H))); % dB scale
title('Magnitude of Frequency Response (dB) - Band-pass');
xlabel('Frequency ( $\pi$  units)');
ylabel('Magnitude (dB)');

% Impulse response
h = impz(b, a, 30); % First 30 samples
subplot(3, 1, 3);
stem(0:29, h, '.');
title('Impulse Response - Band-pass');
xlabel('n');
ylabel('h(n)');

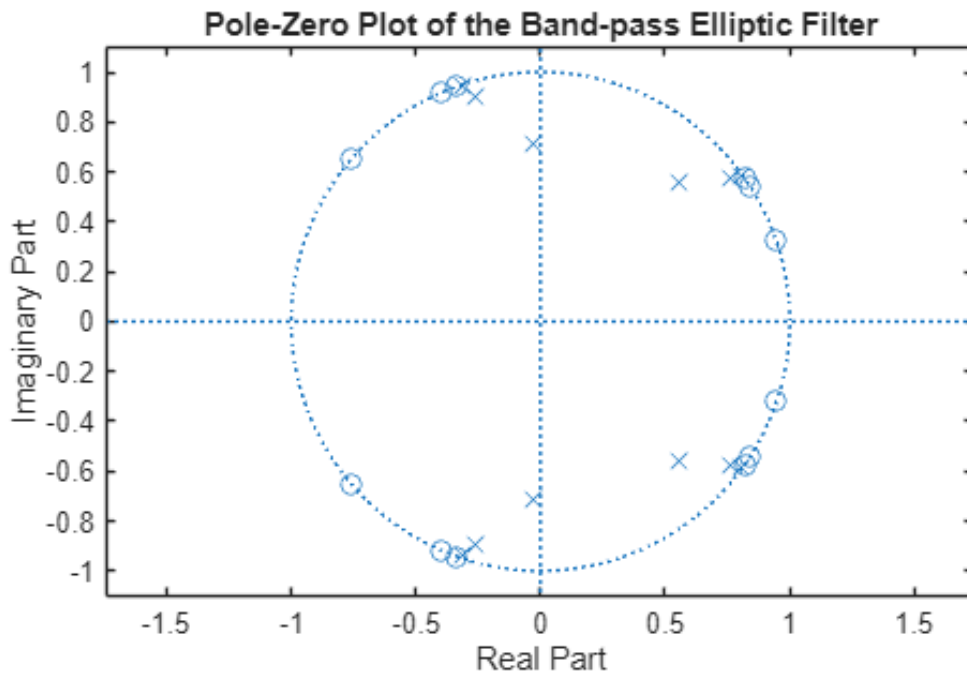
```



```

% Pole-Zero Plot
figure;
zplane(b, a);
title('Pole-Zero Plot of the Band-pass Elliptic Filter');

```

```
% Band-Pass Elliptic Filter
disp('Band-Pass Elliptic Filter:');
```

Band-Pass Elliptic Filter:

```
disp(' ');
```

```
disp('1. Magnitude Frequency Response:');
```

1. Magnitude Frequency Response:

```
disp('The response shows the pass-band between  $0.2\pi$  and  $0.6\pi$ , and the roll-off is sharp on both sides of this peak.');
```

The response shows the pass-band between 0.2π and 0.6π , and the roll-off is sharp on both sides of this peak.

```
disp(' ');
```

```
disp('2. Impulse Response:');
```

2. Impulse Response:

```
disp('The impulse response looks oscillatory, which makes sense because a band-pass filter allows certain frequencies to pass through.');
```

The impulse response looks oscillatory, which makes sense because a band-pass filter allows certain frequencies to pass through.

```
disp(' ');
```

```
disp('3. Pole-Zero Plot:');
```

3. Pole-Zero Plot:

```
disp('The poles are concentrated around the center frequency, and the zeros are placed on either side to attenuate higher and lower frequencies.');
```

The poles are concentrated around the center frequency, and the zeros are placed on either side to attenuate higher

```
disp(' ');
```

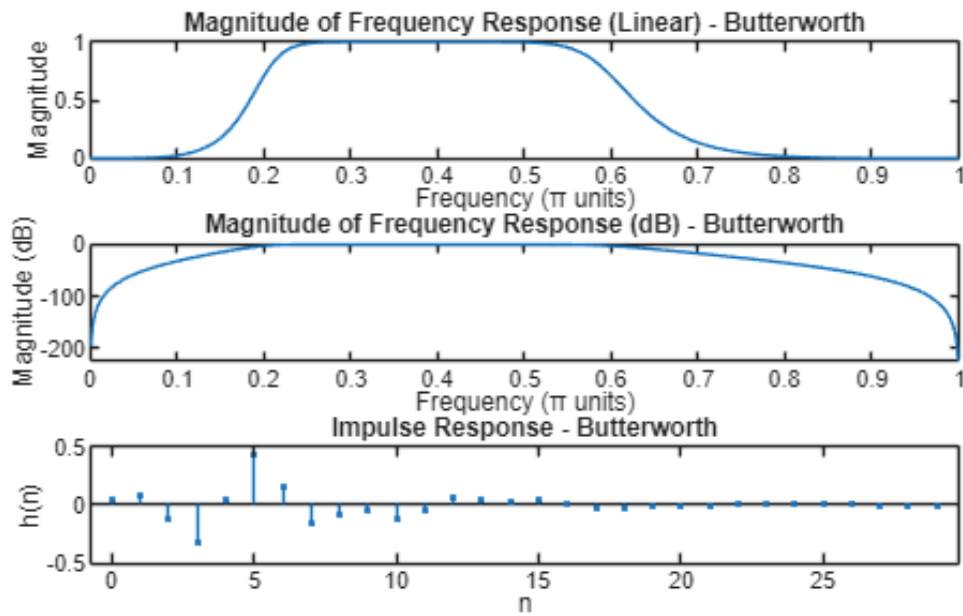
5. Butterworth Filter

```
% Butterworth low-pass filter design
[b, a] = butter(4, Wn); % Order 4, cutoff frequency Wn

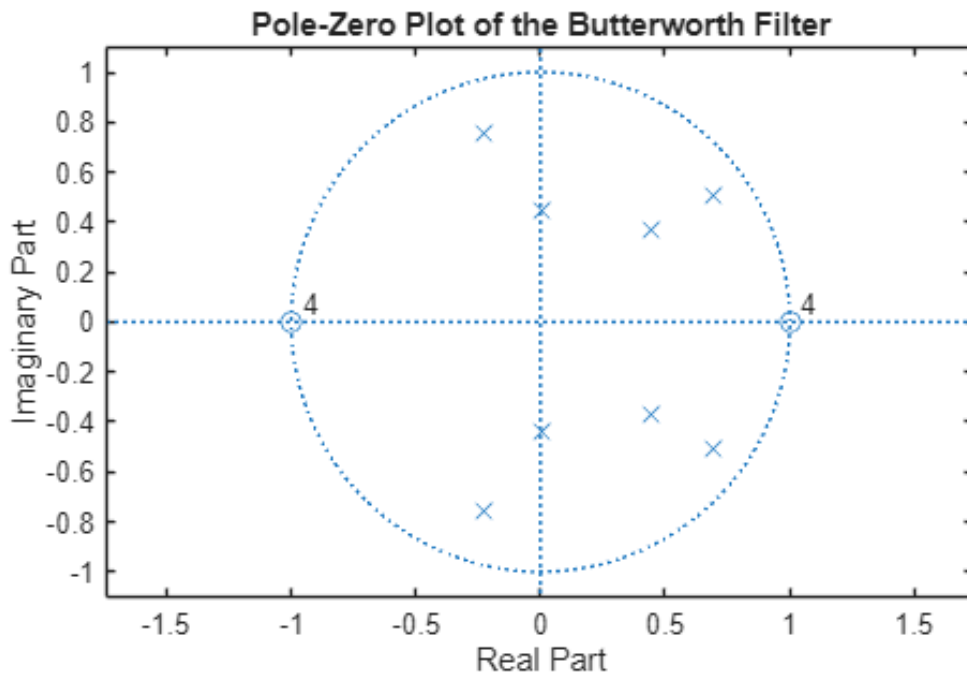
% Frequency response (magnitude)
[H, w] = freqz(b, a, 1024);
figure;
subplot(3, 1, 1);
plot(w/pi, abs(H)); % Linear scale
title('Magnitude of Frequency Response (Linear) - Butterworth');
xlabel('Frequency ( $\pi$  units)');
ylabel('Magnitude');

subplot(3, 1, 2);
plot(w/pi, 20*log10(abs(H))); % dB scale
title('Magnitude of Frequency Response (dB) - Butterworth');
xlabel('Frequency ( $\pi$  units)');
ylabel('Magnitude (dB)');

% Impulse response
h = impz(b, a, 30); % First 30 samples
subplot(3, 1, 3);
stem(0:29, h, '.');
title('Impulse Response - Butterworth');
xlabel('n');
ylabel('h(n)');
```



```
% Pole-Zero Plot
figure;
zplane(b, a);
title('Pole-Zero Plot of the Butterworth Filter');
```



```
% Butterworth Filter (4th Order)
disp('Butterworth Filter (4th Order):');
```

```
Butterworth Filter (4th Order):
```

```
disp(' ');
```

```
disp('1. Magnitude Frequency Response:');
```

1. Magnitude Frequency Response:

```
disp('The roll-off is smooth with no ripples in the pass-band. The transition from  
pass-band to stop-band is slower than the elliptic filter.');
```

The roll-off is smooth with no ripples in the pass-band. The transition from pass-band to stop-band is slower than the elliptic filter.

```
disp(' ');
```

```
disp('2. Impulse Response:');
```

2. Impulse Response:

```
disp('The impulse response decays smoothly, with no oscillations or sharp spikes.');
```

The impulse response decays smoothly, with no oscillations or sharp spikes.

```
disp(' ');
```

```
disp('3. Pole-Zero Plot:');
```

3. Pole-Zero Plot:

```
disp('The poles scattered across the interior of the z-plane, almost evenly  
distributed so there are no ripples in the pass-band. The zeros are placed at 0  
and at  $\pi$  to prevent oscillations.');
```

The poles scattered across the interior of the z-plane, almost evenly distributed so there are no ripples in the pass-band.

```
disp(' ');
```

6. Chebyshev I Filter

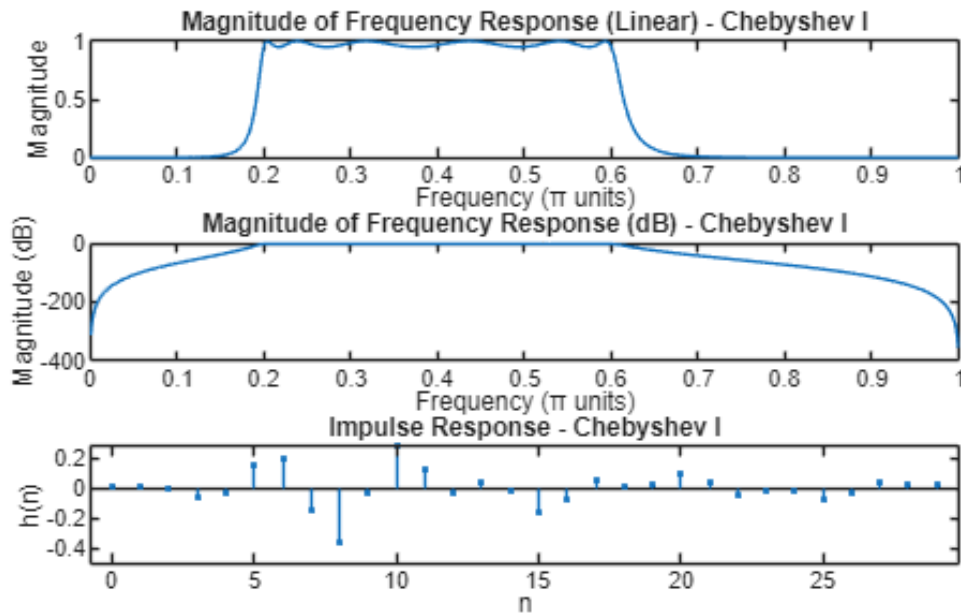
```
% Chebyshev I low-pass filter design  
[b, a] = cheby1(6, Rp, Wn); % Order 6, passband ripple Rp  
  
% Frequency response (magnitude)  
[H, w] = freqz(b, a, 1024);  
figure;  
subplot(3, 1, 1);  
plot(w/pi, abs(H)); % Linear scale  
title('Magnitude of Frequency Response (Linear) - Chebyshev I');  
xlabel('Frequency ( $\pi$  units)');  
ylabel('Magnitude');  
  
subplot(3, 1, 2);  
plot(w/pi, 20*log10(abs(H))); % dB scale
```

```

title('Magnitude of Frequency Response (dB) - Chebyshev I');
xlabel('Frequency ( $\pi$  units)');
ylabel('Magnitude (dB)');

% Impulse response
h = impz(b, a, 30); % First 30 samples
subplot(3, 1, 3);
stem(0:29, h, '.');
title('Impulse Response - Chebyshev I');
xlabel('n');
ylabel('h(n)');

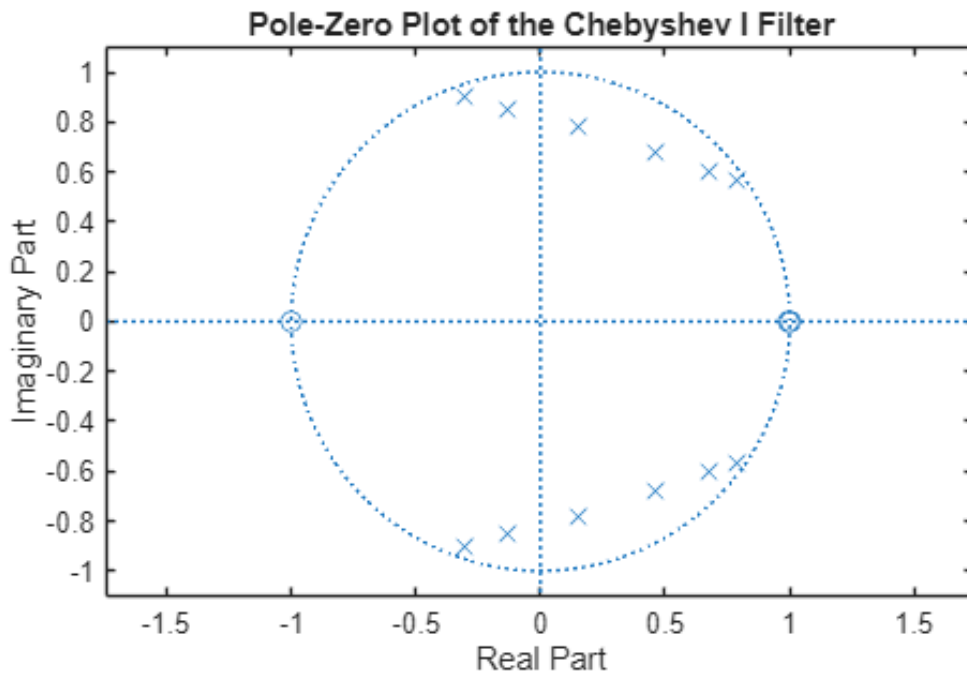
```



```

% Pole-Zero Plot
figure;
zplane(b, a);
title('Pole-Zero Plot of the Chebyshev I Filter');

```



```
% Chebyshev-I Filter (6th Order)
disp('Chebyshev-I Filter (6th Order):');
```

Chebyshev-I Filter (6th Order):

```
disp(' ');
```

```
disp('1. Magnitude Frequency Response:');
```

1. Magnitude Frequency Response:

```
disp('The pass-band has noticeable ripples, and the roll-off is steeper compared to
the Butterworth filter.');
```

The pass-band has noticeable ripples, and the roll-off is steeper compared to the Butterworth filter.

```
disp(' ');
```

```
disp('2. Impulse Response:');
```

2. Impulse Response:

```
disp('There are more oscillations in the impulse response because of the ripples in
the pass-band.');
```

There are more oscillations in the impulse response because of the ripples in the pass-band.

```
disp(' ');
```

```
disp('3. Pole-Zero Plot:');
```

3. Pole-Zero Plot:

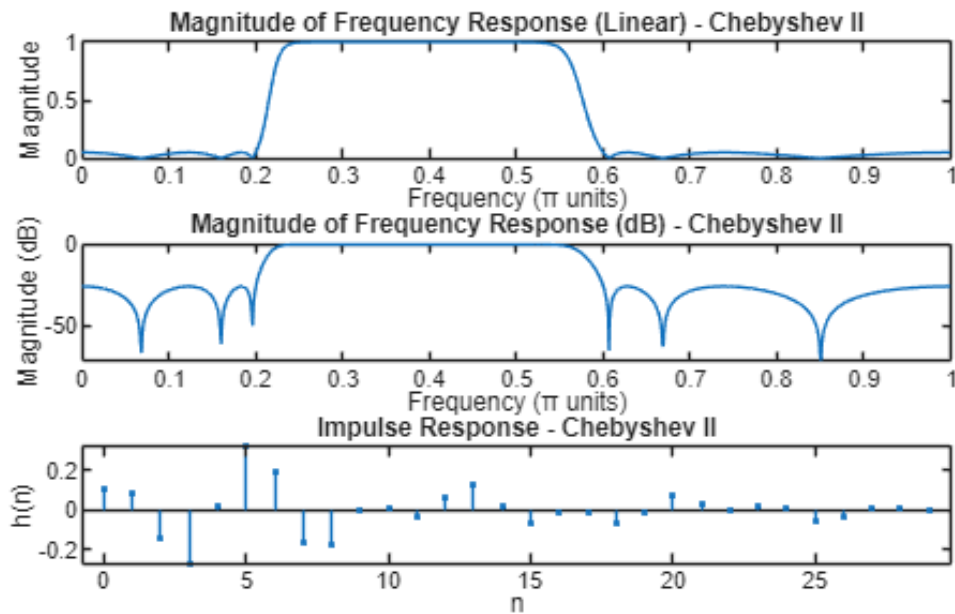
```
disp('The poles are very close to the unit circle, and the zeros are placed at 0  
and at  $\pi$  to prevent oscillations during attenuation.');
```

The poles are very close to the unit circle, and the zeros are placed at 0 and at π to prevent oscillations during attenuation.

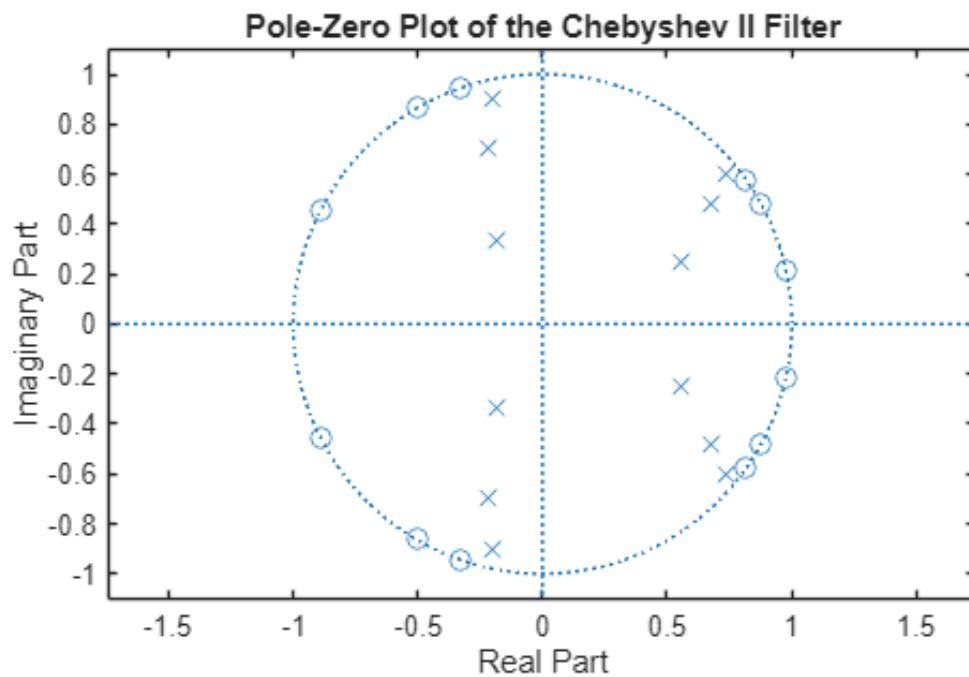
```
disp(' ');
```

7. Chebyshev II Filter

```
% Chebyshev II low-pass filter design  
[b, a] = cheby2(6, Rs, Wn); % Order 6, stopband ripple Rs  
  
% Frequency response (magnitude)  
[H, w] = freqz(b, a, 1024);  
figure;  
subplot(3, 1, 1);  
plot(w/pi, abs(H)); % Linear scale  
title('Magnitude of Frequency Response (Linear) - Chebyshev II');  
xlabel('Frequency ( $\pi$  units)');  
ylabel('Magnitude');  
  
subplot(3, 1, 2);  
plot(w/pi, 20*log10(abs(H))); % dB scale  
title('Magnitude of Frequency Response (dB) - Chebyshev II');  
xlabel('Frequency ( $\pi$  units)');  
ylabel('Magnitude (dB)');  
  
% Impulse response  
h = impz(b, a, 30); % First 30 samples  
subplot(3, 1, 3);  
stem(0:29, h, '.');  
title('Impulse Response - Chebyshev II');  
xlabel('n');  
ylabel('h(n)');
```



```
% Pole-Zero Plot
figure;
zplane(b, a);
title('Pole-Zero Plot of the Chebyshev II Filter');
```



```
% Chebyshev-II Filter (6th Order)
disp('Chebyshev-II Filter (6th Order):');
```

```
Chebyshev-II Filter (6th Order):
```

```
disp(' ');
```



```
disp('1. Magnitude Frequency Response:');
```

1. Magnitude Frequency Response:

```
disp('The pass-band is smooth with no ripples, but there are noticeable ripples in the stop-band.');
```

The pass-band is smooth with no ripples, but there are noticeable ripples in the stop-band.

```
disp(' ');
```

```
disp('2. Impulse Response:');
```

2. Impulse Response:

```
disp('The impulse response decays smoothly with fewer oscillations compared to Chebyshev-I.');
```

The impulse response decays smoothly with fewer oscillations compared to Chebyshev-I.

```
disp(' ');
```

```
disp('3. Pole-Zero Plot:');
```

3. Pole-Zero Plot:

```
disp('Similar to Battenworth, the poles are distributed so that there is no ripple in the pass band, but the zeros are placed differently so that there are ripples during attenuation.');
```

Similar to Battenworth, the poles are distributed so that there is no ripple in the pass band, but the zeros are placed differently so that there are ripples during attenuation.

```
disp(' ');
```

```
% General Observations
```

```
disp('General Observations:');
```

General Observations:

```
disp(' ');
```

```
disp('Elliptic Filters have the sharpest frequency response, but you can see the ripples in both the pass-band and stop-band.');
```

Elliptic Filters have the sharpest frequency response, but you can see the ripples in both the pass-band and stop-band.

```
disp('Butterworth Filters are the smoothest, with no ripples in the pass-band, but  
they roll off more gradually.');
```

Butterworth Filters are the smoothest, with no ripples in the pass-band, but they roll off more gradually.

```
disp('Chebyshev-I Filters show ripples in the pass-band but a steeper roll-off than  
Butterworth filters.');
```

Chebyshev-I Filters show ripples in the pass-band but a steeper roll-off than Butterworth filters.

```
disp('Chebyshev-II Filters have no ripples in the pass-band but show ripples in the  
stop-band, and they also have a steep roll-off.');
```

Chebyshev-II Filters have no ripples in the pass-band but show ripples in the stop-band, and they also have a steep